Connecting Wind Power Plant with Weak Grid – Challenges and Solutions

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***Abstract* — The size of individual wind power plant is continuously increasing, while sites with good wind conditions often are located far from electrical loads. This often results in wind power plants connecting to weak transmission grids. The short circuit ratios at the point of common coupling of wind power plant can be lower than 3 in many cases, and even lower than 2 in extreme cases. This paper analyzes the problems of connecting wind power plant with a weak AC system through detailed voltage stability analysis, small signal stability analysis and transient stability analysis, using power flow, frequency domain and time domain simulation methods. Among the technical challenges, the voltage stability is identified as most critical to the stable operation of wind power plant within weak grid. If the wind power plant itself cannot provide sufficiently fast and extensive compensation, the typical solution for the voltage stability problem is to install dynamic reactive power compensation with fast voltage control capability, such as STATCOM or even Synchronous Condenser. Such additions heavily increase investment cost. In this paper, a coordinated control method of wind power plant is proposed, to minimize the size of any additional reactive power compensation, and it is compared to de facto voltage controllers. The new method enables wind power plant to be controlled as an integral generation source to fulfill grid code requirements, configuring individual WTGs to work as stiff voltage sources. However, with the increase of bandwidth of voltage controller and decrease of grid short circuit ratio, the system is susceptible to a shunt resonance between voltage controller and grid impedance, and its influence on the proposed method is discussed.**

Keywords- Wind Turbine Generator, Wind Power Plant, Weak Grid, Weak AC Connection, Voltage Stability, Small Signal Stability, Fault Ride Through

I. Introduction

Nowadays a large number of potential good wind locations are found far away from generation centers in many countries and the wind power plants (WPPs) will thus be integrated into power network through long transmission lines. The external grid may be weak at point of common coupling (PCC), which means low short circuit ratio (SCR). In order to make sure that WPPs are in compliant with grid code requirements, a set of quality and reliability requirements, including operating voltage range, power factor range and fault ride through (FRT), have to be fulfilled. It is often found that the performance of WPP deteriorates with the decrease of grid strength or grid SCR. In order to ensure WPP's stable operation under weak grid connection, the following questions are investigated in this paper

1. What is the lowest SCR of the external grid that a WPP is able to be connected and is able to satisfy grid code requirements?
2. What are the bottlenecks/problems of the system when WPP is connected with weak grid?
3. How can the WPP performance be improved by solving those identified problems?

In [1] the authors were discussing about voltage quality, thermal capability and stability of WPP integrating into a weak grid. However the conclusions were just based on dynamic simulation results of the Norwegian power network when the WTGs worked at their full power output. It did not discuss the minimum SCR that WPP can be connected and how to improve its performance in different situations. In [2] three major problems for WTGs with weak grid connection, namely recovery after a fault, oscillations and active/ reactive power capabilities, were discussed. But it didn't include any quantitative and detailed analyses. The paper [3] described the performance of a WPP under different severe disturbances as well as its recovering period following the fault clearance. It considered the GE 1.5 MW WTG with simplified representation of WTG control structure, together with wind farm management system (WFMS). Similarly, the behavior of system was simulated without addressing on the impacts of external grid SCR. And only dynamic simulation was used to analyze the stability problem, which did not clearly define the stability margin of the WPP. In [4] a stability study of a WTG connected to very weak ac grids with varying grid SCR was presented. The WTG model and its behavior were firstly done in Matlab/Simulink and the verification was then followed up by a detailed model within PSCAD/EMTDC. The main paper's focus was on small signal stability by using eigenvalue analysis to find out the root cause and function blocks with most negative influences in its control scheme. However it doesn't take into account the effect of wind farm control or power plant controller (PPC) that exists in most of the modern WPPs. E. Muljadi and other authors' work [5] showed the positive impacts on system stability of connecting WPP in comparison with conventional generators with weak grid, without discussing the WPP's own stability problems. In [6] a hybrid DC link adaptive control algorithm was proposed for full power converter when it was operated in a weak grid system.

In this paper, the problems of connecting wind power plant with a weak AC system are studied through detailed voltage stability analysis, small signal stability analysis and transient stability analysis, using power flow, frequency domain and time domain simulation methods. In order to improve the stability of WPP connected with weak grid, while at the same time minimizing the size of any additional reactive power compensation, a coordinated control method of wind power plant is proposed, and it is compared to de facto voltage controllers. The new method enables wind power plant to be

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controlled as an integral generation source to fulfill grid code requirements, configuring individual WTGs to work as stiff voltage sources.

II. Operating WPP in a weak grid connection in steady state

As per grid code requirements, WPP must have certain voltage control, reactive power and/or power factor control capabilities. With the increase of grid impedance or decrease of grid SCR, the WPP's reactive power capability has to be investigated to see if it is large enough to meet the grid code, while at the same time it can still keep the nodes' voltage with in their constraints. In this section, cases with and without considerations of main WPP transformer's on load tap changing (OLTC) are studied.

At steady state study it is important to examine the voltage level at buses within WPP based on their specifications. In general voltages at PCC, HV and LV of WPP and WTG transformers and terminal voltage of WTGs are considered. Furthermore the voltage at the inverter side should be kept in range as well because of the limitation of available DC voltage.

A. Base case study

For power flow study the WPP scheme without compensation is simplified to the base case with single WTG. From the voltage point of view this simplification should be done in such a way that the obtained case reflects proper voltage drops from the grid connected point. In general the grid is modeled as a constant voltage source (an infinite/slack bus) behind impedance.

As another assumption the impedances of WPP collector system between the WTGs are neglected, as it is very small compared with the large grid impedance for weak grid connection of WPP. In that case the voltages at WTG terminal are more or less similar in the WPP so that the behaviors of all WTGs are similar. It means all WTGs can be presented as a single WTG connecting to the WPP medium voltage bus bar through one medium voltage (MV) cable.

The tap ratio of WTG and WPP may be changed during operation. The detailed equivalent WPP configuration is shown in Fig. 1 where the π- scheme of cable is applied for MV cable equivalent. The nodes' voltages are marked as: $V_{WTG,INV}$ - voltage at grid side inverter terminal; $V_{WTG,LV}$ - voltage at LV of WTG transformer; V_{PCC} - voltage at PCC of WPP.

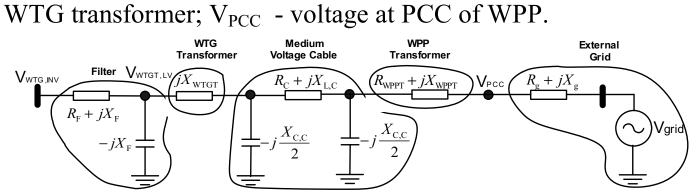

Figure 1: Detail Equivalent WPP Scheme with MV cable.

B. Allocation operating points of WPP in a weak grid connection

Since the WTG terminal voltage must be in range [0.9 1.1] due to relay protection setting as stated in general specification and the WTG inverter terminal voltage should be in range [0.9 1.143] in order to adopt linear modulation index with the limited DC voltage, the following constraints should be satisfied

$$0.9 \le V_{\mathrm{LV,WTGT}} \le 1.1 \qquad (1)$$

$$0.9 \le V_{\mathrm{WTG}} \le 1.143 \qquad (2)$$

The equations (1) and (2) are constraints on LV side of WTG transformer and WTG terminal voltage.

It can be seen that $V_{\mathrm{LV,WTGT}}$ and V_{WTG} depend on the WTG active/reactive power outputs as well as R_g and X_g. Thus with constraints (1) and (2) the allowable transmitted area of WTGs is a function of SCR and XoR.

In modern WPPs, the main power transformers are normally equipped with OLTC transformer that is able to regulate the voltage level at WPP's MV side. It is a very useful method for WPP to meet voltage requirements as per grid code at PCC. However in this case study, equations (1) and (2) cannot guarantee that the PCC voltage will be in the acceptable range with the change of tap ratios. Therefore the influence of OLTC transformer has to be investigated, considering an additional constraint on voltage at PCC

$$V_{\min}^2 \le V_{\mathrm{PCC}}^2 \le V_{\max}^2 \qquad (3)$$

The allocation for WTGs is implemented for the circuit as shown in Fig. 1. using equations (1), (2) and (3). The cable length equal to 2 km is applied. The results showed that the allowable transmitted WTG PQ area is more narrow with smaller SCR in both high and low XoR cases.

The results of a particular case study are presented in Fig. 2.

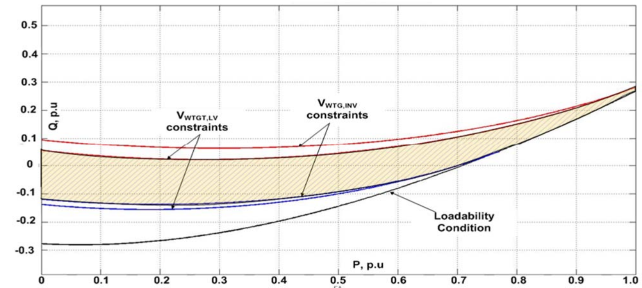

Figure 2: Allowable transmitted area of V112 WTG to external grid model with SCR=1.27 and XoR=4

It should be reminded that the whole x-y area is PQ chart capability of WTG while the dashed areas are its allowable transmitted PQ. It is realized that if SCR is decreased to some lower value the WTG terminal voltage will be higher than 1.1 p.u when WTG outputs 1.0 p.u active power. This SCR is the critical value for connecting WTGs to external grid. In general it is also a function of external grid voltage and different active power output levels as shown in Fig. 3 and Fig. 4.

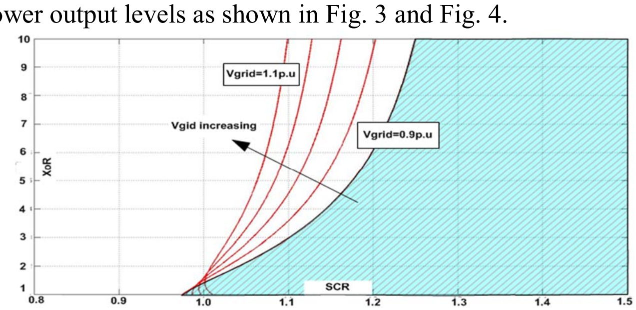

Figure 3: Critical SCR as a function of XoR with different external grid voltages

In Fig. 3 and Fig. 4 the dashed areas showed the conditions of external grid to which the WTG can be connected. It can be observed that the allowable (SCR, XoR) area is extended in the left direction with higher external grid voltage and lower WTG active power output which is better from the connecting capability point of view. The explanation can be found in the larger voltage drop in case of bigger WTG active power output so that with higher voltage of external grid voltage magnitudes of other network buses are in acceptable range.

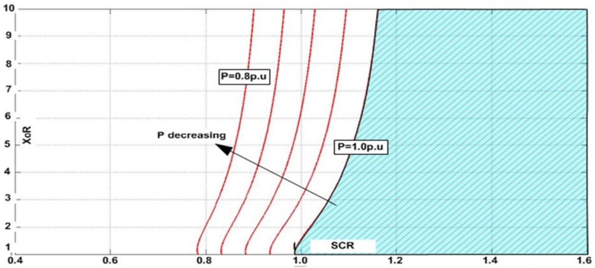

Figure 4: Critical SCR as a function of XoR with different active power outputs

If the WPP is equipped with OLTC main transformer, the OLTC transformer is able to regulate voltage. It therefore widens the transmitted PQ area (Fig. 5) and consequently, the allowable area (SCR, XoR) in that WTG can be connected to external grid.

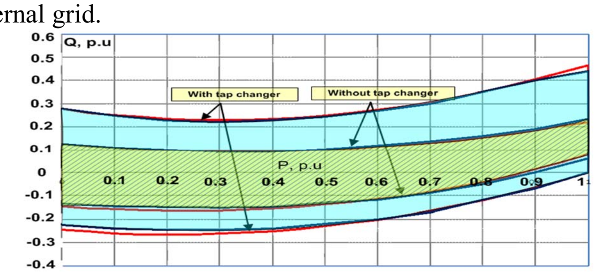

Figure 5: Impact of OLTC on the allowable transmitted PQ from WTG to external grid, SCR=2.5, XoR=4.0

C. Voltage stability analysis

Voltage stability analysis is carried out based on the PV, QV curves with different SCR and XoR of external grid. The PV curves are presented as shown in Fig. 6 with SCR varying from 1.2 to 2.8, while the XoR is fixed at 5.

In Fig. 6 it can be seen that the margin of WTG active power output decreases when grid strength or grid SCR decreases. Results show that in order to transmit 1.0 p.u active power and zero reactive power output, in order to keep voltage within its range [0.9 1.1] p.u., the grid SCR should be larger than 2.5. The figure represent the voltage stability problem when WPP is connected with weak grid, i.e., the reactive power support should be large enough to keep the voltage within the acceptable range, when transmitting active power to the grid.

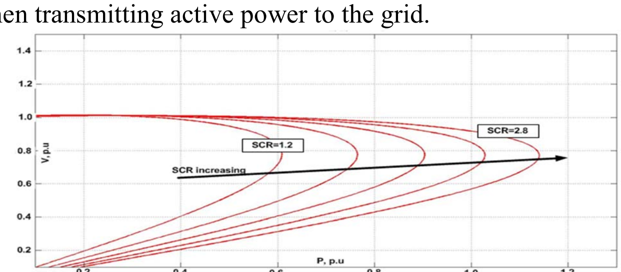

Figure 6: PV curve at WTG terminal with SCR variation with XoR=5 and Q=0 p.u.

The OLTC has a positive impact on WPP reactive capability and thus the necessary additional reactive power compensation for WPP to meet grid code requirement is smaller in case of OLTC installed.

III. Small signal stability analysis for WPP in a weak grid connection

The objective of small signal stability analysis is to investigate the WPP response under small disturbances. In this context a disturbance is considered small if the equations that describe the resulting response of the system can be linearized for the purpose of analysis.

A. Wind turbine generator

A Simulink model is created to present the grid model together with control structure of WTG full converter type. Together with WTG control structure the overview of model is represented as given in Fig. 7, while the turbine's mechanical system, generator and generator side converter and controller are omitted.

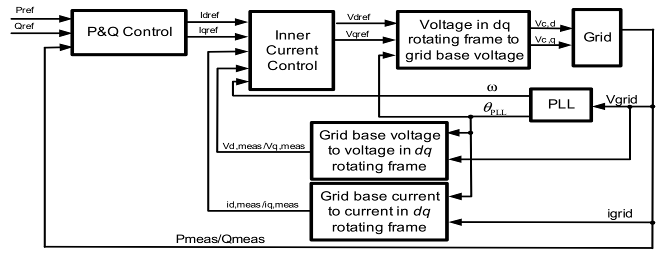

Figure 7: Current control structure of WTG

In the small signal model, the DC link model, Vdc and Q control, Inner Current Control (ICC), grid model, phase locked loop block (PLL) and a number of typical low pass filters and grid interfaces are included. Their detailed controller scheme can be found in many papers and monographs and thus are not described here. Their control parameters are same as real the WTG.

B. Power plant control

PPC aims to control active power output and the voltage or power factor at PCC as per grid code requirements. At steady state operating conditions, PPC sends active and reactive power reference to WTG controller through communication channels. In stability analysis the time delay of the communication must be considered. If the compensation devices are installed in WPP then they receive the reactive power set points from PPC as well. During fault ride through (FRT) periods the PPC is frozen and the WTG works in advanced grid option (AGO) mode.

The reactive power control loop within PPC is either PI or droop controllers.

C. Grid model

A detailed model of grid in p.u system base is shown in Fig. 1. The model consists of WTG filter, WTG transformer, medium voltage cable, WPP transformer and external grid.

The parameters of WTG filter, WTG transformer are often given in manufacture's specification while other parameters are varied up to WPP design and external grid. Particularly, R_g and X_g are determined according to external grid SCR and XoR.

D. Results of small signal stability study

Firstly, the WPP without PPC is applied in this case study. The root locus of the system is drawn when the SCR of external grid varies from 1.5 to 5.5 with XoR=10 (Fig. 8).

It can be seen that the system is more stable with stronger external grid, i.e, with higher SCR. For example system is not stable at SCR=1.5 when a small disturbance occurs.

The conducted eigenvalue analysis proved that the PLL has the largest impact on the dominant modes.

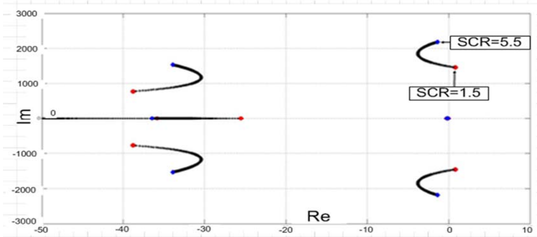

Figure 8: Root locus for varying SCR from 1.5 to 5.5 with XoR=10

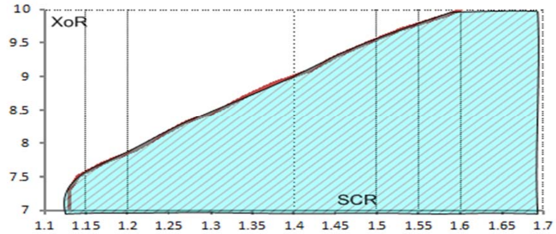

Figure 9: The acceptable SCR area as a function of XoR in small signal stability analysis

The similar study on the impacts of PI parameters of DC link voltage controller, reactive power controller, inner current controller and PLL are also carried out. The root locus of the system is also depicted when the XoR of external grid varies. Based on obtained results it is concluded that in order to bring system from unstable region to stable region, PI control parameters of DC link control block, PLL control block, voltage control block and communication time delay should be decreased, PI parameters of reactive power control block should be increased while PI parameters of inner current control should be optimized at some point in the middle.

The study also showed that with current PPC and WTG control coordination structure the small signal stability of the system is not improved. The allowable grid condition (SCR, XoR) that a WTG can be connected to is given in Fig. 9.

IV. TRANSIENT STABILITY ANALYSIS OF WPP IN A WEAK GRID CONNECTION

The previous discussions on the voltage stability and small signal stability only represent the linear behavior of WPP connecting with a weak grid. However as the WTG or WPP control system is nonlinear under large disturbance, dynamic simulation is carried out to analyze the transient stability of WPP under large disturbance.

In this section, the dynamic simulation results for the WPP connected with weak grid are presented. The aim is to analyze the transient stability of a WPP with current WTG and PPC under large grid disturbances, without additional reactive power support from other equipment such as STACOM or synchronous condenser. The simulations are conducted on the PSCAD software platform.

The dynamic simulations are conducted at SCR equal to 1.6, 1.8 and 2 by using a base case system, same as in Section II.A.

The grid faults simulated are severe voltage dip and shallow voltage dip. The results show that with the current WTG controller and PPC:

- When the SCR is lower than 2, the base case system might lose stability under large disturbances or faults, which are related with voltage stability problems. WTGs are either tripping off due to over voltage at fault recovery period or repeatedly entering and leaving LVRT modes.
- Either the WTG/WPP can provide fast enough voltage control capability, or additional equipment with fast voltage control capability, such as STATCOM or Synchronous Condenser, should be installed.

The stability problem that WTG repeatedly entering and leaving LVRT modes is analyzed here. The PPC is frozen until all WTGs are out of LVRT mode, and it is noticed that not all WTGs step out of LVRT modes simultaneously. This means that when PPC is still freezing, some WTGs step out their LVRT modes, and regain their PQ control modes. However their Q references are still the values pre-fault sent by the PPC, and they are not updated due to the WTGs' output active power and also grid strength. Furthermore, the grid impedance may decrease significantly due to tripping one or more transmission line. This is not a big issue for strong grid, however for weak grid the reactive power reference Q_ref must be aligned with the active power production and the grid strength, other severe voltage oscillation will happen. The above analysis explains that in a very weak grid system, the WTG needs fast voltage control capability after the grid fault is cleared.

With the above voltage stability, small signal stability and transient stability analysis, the potential problems of WPP connecting with weak grid are identified. In the next section, a coordinated voltage control strategy of WPP and WTG is proposed to solve those problems, and its performance is compared with the current control strategy.

V. PROPOSAL OF COORDINATED CONTROL OF WPP AND WTG FOR A WEAK GRID CONNECTION

As stated in previous sections it is concluded that

- Voltage stability of wind power plant with weak grid connection is a severe problem, which requires fast and accurate reactive power support from wind power plant. Otherwise balance of plant (BoP) component with fast voltage control capability has to be installed.
- During LVRT, some grid codes require large reactive current injection. The excessive large reactive current can cause temporary over voltage after the grid fault is cleared and finally lead to action of over-voltage protection to trip the WTG, if the WTG is connected with a weak grid.
- After LVRT, there is a coordination problem between PPC and WTG. WTGs receive wrong reactive power references, cause large voltage oscillations, and are forced to continuously to enter and leave LVRT modes.

Because of the above listed problems, the WTG/WPP may experience very large voltage oscillation and become unstable and retrigger the FRT modes continuously. To solve this problem, it is proposed to design a fast local WTG voltage controller together with a proper coordination with PPC voltage/reactive power controller. The proposed coordinated control strategy is shown as Fig. 10.

Besides, inappropriate control setting can also cause malfunction of PLL, and lead to WTG tripping by over-frequency protection. This is solved by choosing the appropriate control parameter and PLL settings.

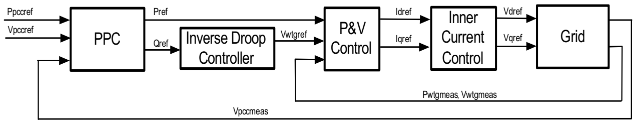

Figure 10: Proposed coordinated control structure of slow WPP voltage/reactive power controller and fast local WTG voltage controller

Two cases are studied to compare the WPP's performance with de facto and the proposed new control structure of PPC and WTG.

- Performances of de facto PPC and WTG control structure and proposed coordinated control structure under disturbance of grid load rejection 0.2pu, as shown in Fig. 11.
- Performances of de facto PPC and WTG control structure and proposed new coordinated control structure under disturbance of sudden grid SCR decreasing from 2.9 to 1.8, as shown in Fig. 12.

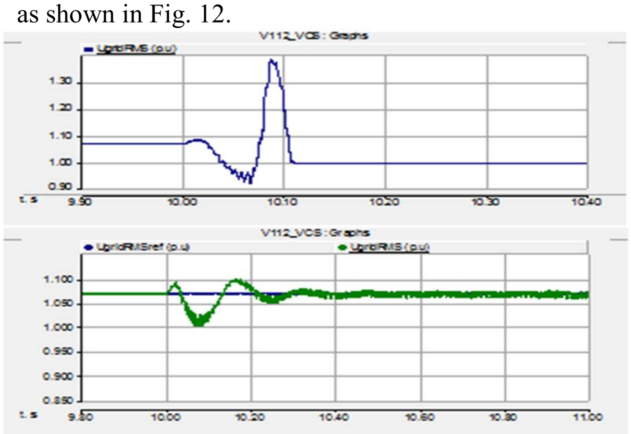

Figure 11: Voltage of WPP with de facto and new proposed PPC and WTG controllers under disturbance of 0.2p.u. load rejection, connecting to external grid with SCR=1.8 and XoR=10. Upper graph: de factor controller; bottom graph: new proposed controller

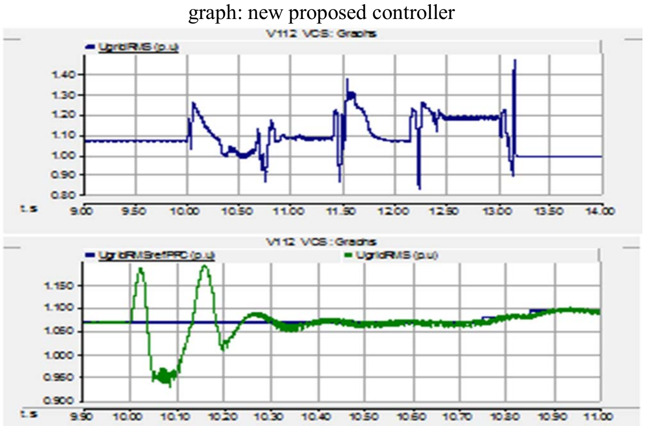

Figure 12: Voltage of WPP with de facto and proposed new PPC and WTG controllers, under disturbance of sudden grid SCR change, grid SCR decreased from 2.9 to 1.8. Upper graph: de factor controller; bottom graph: new proposed controller

It is shown that the new proposed coordinated PPC and WTG voltage controller improve the WPP's performance significantly under both disturbances, while WTGs of the WPP equipped with de facto PPC and WTG controller are tripped under the same disturbances. The WTGs are tripped by their over-voltage protections, because the too slow voltage regulation capability of de facto PPC controller cannot control the voltage fast enough within the allowable range.

However, with the increase of bandwidth of fast local voltage controller and decrease of grid SCR, the system is susceptible to a shunt resonance interaction between voltage controller and grid. This is similar as the series resonance between current controller and grid which is analyzed in [7]. The shunt resonance mode will destabilize the voltage controller, which requires either decreasing the bandwidth of voltage controller or modification of the control block. This might cause the failure of the implementation of the fast WTG voltage controller and will be investigated in the future work.

VI. CONCLUSION

In this paper the challenges of operating WPP with a weak grid connection in normal and grid faulted conditions are addressed.

Power flow study shows the allowable operating range of WTG shrinks when external grid's SCR decreases, and it is found that the voltage stability is the most severe problem. The small signal stability analysis points out that system starts to loss stability as the grid SCR decreases to a certain value. The PLL controller has the most negative impact on the dominant system modes. The allowable operating range of a WPP connected to the external grid with different grid SCR and XoR are shown in Fig. 3, Fig. 4 and Fig. 9. Furthermore, dynamic simulations of a WPP under different types of grid faults are also carried out to study the WPP's nonlinear behavior under large disturbance. Several problems are identified and their root causes are analyzed.

As a solution for the above problems, an improved coordinated control strategy combining slow WPP voltage/reactive power controller and fast local WTG voltage controller is proposed in this paper. This new method enables wind power plant to be controlled as an integral generation source to fulfill grid code requirements, configuring individual WTGs to work as stiff voltage sources. The advantages of the proposed method are discussed and presented through simulation results. However, with the increase of bandwidth of voltage controller and decrease of grid short circuit ratio, the system is susceptible to a shunt resonance between voltage controller and grid impedance, and its influence on the proposed method is discussed.

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